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Patents Revisited: Their Function in Developing Countries

*By Constantine Vaitsos**

The patent system in developing countries has a predominantly negative effect and is devoid of significant benefits for these countries; virtually all owned by large foreign corporations, patents are used as a vehicle for achieving monopoly privileges which militate against conditions conducive to foreign investment, and hinder the flow of technology to developing countries while restricting their technological advance through imitation and adaptation. For developing countries to license patents is therefore tantamount to granting import permits under restrictive conditions.

'Patent systems are not created in the interest of the inventor but in the interest of national economy. The rules and regulations of the patent systems are not governed by civil or common law, but by political economy' [*Government of India, 1959, para. 20*].

'Except, perhaps, cases of warranty of horses, there was no subject which offered so many opportunities for sharp practice as the law of patents' —House of Lords, Parliamentary Debates, July 1, 1851, col. 18.

'Today, United States policy with respect to international patent relations must be shaped in the light of vital, even if at times confused, foreign policy objectives' [*U.S. Senate, 1957, p. 4*].

INTRODUCTION

The 'economic theories', the arguments and the rationale behind the presently-existing patent system have been developed almost exclusively by patent lawyers and jurists supported by business executives. Thoughtful economic analysis has been quite absent.[1] In its place broad generalizations, based mostly on legal grounds—though some are theological, such as the principle of 'natural rights' of inventors—underlie the presently accepted notions about industrial property privileges The 19th century debate on the patent system which came close to abolishing patents in England and Germany and actually temporarily abolished them in other countries, marked the end of the active participation by economists in the subject. The absence of serious economic analysis is surprising, given that patents are often a powerful economic tool. They may determine or sometimes hinder the possibilities of further technological advancement, influence or retard the timing of commercial introduction of inventions (for example, in the case of fluorescent lighting, methyl methacrilate

* An earlier and longer version of this paper was presented to the Junta of the Andean Pact as a background work for its proposal on industrial property legislation.

plastics and cement carbide). They may also underpin particular structures of technological or commercial control in certain sectors (e.g. as in synthetic rubber, or in some types of synthetic fibres). Sometimes patents might determine the world-wide dominance of a whole branch of an industry by certain firms (this is the case at present in the pharmaceutical industry; an earlier example is the classical cartel case of the 'Europaischer Verband der Flaschenfabriken' involving an automatic machine for making glass bottles).

This paper will attempt to evaluate some of the major economic issues and implications resulting from the patent system as it functions at present in developing countries. We will not deal just with the problem of 'abuses' like restrictive business practices or excessive monopoly privileges. Such a limitation is usually a way of circumventing debates on patents themselves and, implicitly assumes the acceptance *per se* of the underlying principles of the present patent system. Instead we will attempt to analyse the function and overall effect of patents on economies of developing countries.

In those discussions on patents which are influenced by legal analysis the subject appears quite complex and seems to require in-depth knowledge by specialists. Various intricate legal specifications,[2] complex administrative and procedural requirements for the application of the patent laws as well as a considerable degree of arbitrariness in defining, for example, what is sufficiently 'novel' or industrially 'util' to be patented, lead to a certain confusion as to what patents are and what effects they actually have. The economic principle of patents, though, is clear. The patent is a monopoly grant. This privilege is granted, in developed and in developing countries, on the long-held assumption that patents are a *necessary* incentive for inventive activity and/or are compensation for this activity. Also, through patent disclosure, or the guarantee of monopoly, etc., it is assumed that *sufficient* incentives are given so as to introduce innovations into commercially beneficial industrial activities.[3] *Furthermore*, for the economy of a country as a whole (leaving distributional effects aside) it is assumed that the monopoly costs to consumers or to other producers are smaller than the benefits that accrue from promoting inventive and investment activities through patents. It is important to distinguish that the argument involved here is not with respect to inventions and investments *per se*, but with respect to the role of monopoly privileges on such activities. Monopoly privileges granted by patents clearly intend to bring the production of inventions into a market price framework. Prices exist when scarcity exists. Patents, granting monopoly of usage (or usage under licence), created scarcity by limiting the availability of inventions, although an invention is by its nature an 'inexhaustible' entity in terms of number or times of usage.[4] To a certain extent, prices are put on the usage of inventions not because of their scarcity but *in order* to make them scarce to possible users. A patent diminishes the possible usage of an innovation with the intent (or so as) to generate an economic rent [*Plant, 1934, p. 31; Penrose, 1951, p. 29*]. The monopoly effects of patent-grants on inventive activity should be evaluated not only on a national basis but also on a world-wide one. What might or might not be true of the effect of patents on a particular country's inventive activity need not be so for the effects on similar activities in other countries. In the form that the present international system of patents is set up[5] there are strong reasons to believe that they might definitely hinder inventive activities in some countries (especially the developing ones[6]) even if for others, as some claim, the effect is positive. A study prepared for the United States Senate Judiciary Committee concluded that the provisions of the international patent system '. . . it is evident, have altered the complexion of the patent grant from one designed primarily to stimulate domestic industry to one in which the foreign patentee has an increased chance of producing where he chooses and retaining his patent monopoly' [*United States Senate, 1957, p. 3*].[7] Thus, if patents constitute a necessary incentive to inventive activity (an issue that still remains to be proven) it is necessary to ask: patents granted by whom and for whose inventive activity?[8] We have now to consider the characteristics and effects of patents on the economies of developing countries.

TWO BASIC CHARACTERISTICS OF PATENTS IN DEVELOPING COUNTRIES

1. *Foreign ownership of patents*

One important fact that needs to be stated is that when one talks about patents in developing countries one really means patents owned almost in their totality by foreign companies or foreign nationals. The following table gives some comparative data:

TABLE 1

PATENTS GRANTED TO FOREIGNERS AS A PERCENTAGE OF TOTAL PATENTS GRANTED BETWEEN 1957 AND 1961 INCLUSIVE

'Large' Industrial Countries	%	*'Smaller' Industrialized Countries*	%	*Developing Countries*	%
U.S.A.	15·72	Italy	62·85	India	89·38
Japan	34·02	Switzerland	64·88	Turkey	91·73
West Germany	37·14	Sweden	69·30	United Arab Republic	93·01
United Kingdom	47·00	Netherlands	69·83	Trinidad & Tobago	94·18
France	59·36	Luxemburg	80·48	Pakistan	95·75
		Belgium	85·55	Ireland	96·51

Source: Statistical information appearing in United Nations [*1964, pp. 94–95*].

In fact, Table 1 considerably *overstates* the importance of patents owned by *nationals* of developing countries. The majority of these national patents are owned by individual inventors (not by companies), and have minimal industrial or business importance. For example, recent investigations in Chile's Industrial Property Office indicated that the majority of nationally owned patents belonged to individuals, not to business firms, and their commercial importance 'appeared nil' [*C.O.R.F.O., 1970, p. 16*]. Similar conclusions were reached from research undertaken in Argentina and India [*Government of India, 1959, paras. 25–27*]. Three-quarters of the patents belonging to Argentinans in the late 1960s were owned by individuals. Depth interviews of a sample of Argentinian inventors indicated that 75 per cent of them had never obtained successful results from the industrialization of their discoveries. Of forty inventors only one considered his principal source of income to be his patented and unpatented inventive activities [*Chudnovsky and Katz, 1970, p. 36*].[9] *Thus, if the num-*

ber of patents is weighted by their economic or technological worth (i.e. volume of sales or value added) most developing countries are likely to find that the (weighted) patents belonging to their own nationals amount to a fraction of 1 per cent of the total patents granted by such countries. Thus, when we talk about patents in developing countries we really mean *foreign* patent holdings.

The trend for developing nations indicates that the situation is worsening rather than improving. Large industrialized countries have experienced comparatively insignificant changes in the percentage of their patent grants in national hands. See, for example, Table 2:

TABLE 2

FOREIGN HOLDINGS OF DOMESTIC PATENTS IN LARGE INDUSTRIALIZED COUNTRIES IN APPROXIMATE PERCENTAGE FIGURES

Countries	*1940*[a]	*1957–61*[b]
	%	%
United States	10	16
Japan	25	34
Germany	25	37
United Kingdom	50	47
France	50	59

Source: (a) Diegger [*1948, p. 257*]; (b) Data from Table 1 of the present paper.

Contrast the figures presented in Table 2 with what occurred in Chile between 1937 and 1967. Considering as base 100 the number of patents registered in 1937, the following changes were observed in Chile thirty years later:

TABLE 3

	Nationally-Owned Patents	*Foreign-Owned Patents*	*Total Patents*
1937	100	100	100
1967	110	956	664

Source: C.O.R.F.O. [*1970, p. 15*].[10]

The progressive denationalization of Chilean grants is shown in Table 4.

TABLE 4

PERCENTAGES OF PATENTS REGISTERED IN CHILE AND OWNED BY:

	Nationals	*Foreigners*
	%	%
1937	34·5	65·5
1947	20·0	80·0
1958	11·0	89·0
1967	5·5	94·5

Source: C.O.R.F.O., *op. cit.*, p. 15.

TABLE 5

Industry	Percentage of Total Patents in 1967	Percentage of Industrial Patents owned by Nationals	Foreigners
	%	%	%
Agriculture and food products	8·4	6·4	93·6
Mining and metallurgy	12·3	1·5	98·5
Motors and machinery	3·2	20·0	80·0
Chemicals and pharmaceuticals	[40·6]	[1·6]	[98·4]
Textiles	4·6	17·7	84·3
Domestic arts	6·8	14·6	85·4
Electricity and scientific instruments	10·2	6·0	94·0
Construction	9·5	15·2	84·8
Fish industry and transportation	2·8	9·7	90·3
Military industry	1·5	6·2	93·8

Source: C.O.R.F.O., *op. cit.*, pp. 18–25.

Similarly, in Argentina 62 per cent of patents were owned in 1953 by individual inventors, the majority of whom were Argentinians. Only 15 years later, in 1968, the percentage dropped to 25 per cent. The difference was due to patents obtained by foreign corporations [*Chudnovsky and Katz, 1970, p. 1*].[11] The concentration of practically all of the patents of developing countries in foreign hands suggests that one of the major arguments used to justify the existence of the present patent system, namely, the incentive to domestic inventive activity, does not work in these countries. Foreign-owned patents reflect, if anything, 'non-national' inventive activity and obviously have no direct influence on domestic inventiveness. (There are indirect effects, most of them negative, which we will discuss later.) Furthermore, it is reasonable to assume that the granting of patents by Peru, Turkey or Indonesia or even by the whole Third World taken together has insignificant effects on the R & D plans of transnational corporations like Du Pont, Phillips or Mitsubishi, or on governmental spending on science and technology in, say, the United States or France. Consequently, instead of considering the effects of patents granted by developing countries on inventive activity *per se*, we should rather concern ourselves with such questions as whether they promote or deter foreign investment, whether they enhance or restrict technology transfer or whether they affect the terms of trade. The present patent system as an economic policy tool has as little to do with domestic inventive activity as tax exemptions on profit repatriation by foreign firms or the grant of special market privileges to foreign companies. It is probable that tariff and general import policies have more to do with domestic inventiveness than patents.

2. *Patents and Business Policy*

There has been a serious shift in both industrialized and developing countries, in the concept of 'ownership' of invention. At one time an individual inventor might have 'owned' a few patented inventions; nowadays a business corporation might have hundreds or thousands of patents. Such a shift in the ownership (without quotation marks this time) of patents has revolutionized the concept of the patent and changed its implications: it has removed the patent from the world of the inventor to that of the business executive. '. . . [Patent] use is then no longer attuned to the folk ways of science but to those of business. At once [the patent] . . . the supposed spur to technical progress, becomes the instrument of business policy. A host of moves are made to enhance its pecuniary value' [*Hamilton, 1945, p. 587*].

Business policy exercised through patents can have various effects. The concentration of patents in a few enterprises can greatly enhance market control. The concentration of patents in the dominion of large corporations has been called an 'economic evil' [e.g. *Machlup, 1952, p. 284; Stocking and Watkins, 1951, pp. 468–9*] and patent concentration has brought forth official corrective action as a particular case of monopoly power [*Kaysen and Turner, 1959, pp. 165 ff.*]. Since a detailed discussion of the effects of economic concentration on economic development would carry us far from our present topic we will limit ourselves here to citing some indicative statistics on concentration. About 50 per cent of all the patents acquired by private contractors as a result of government-financed research in the United States are owned by 20 large corporations [*Watson and Holman, 1967, p. 381*] (U.S. Federally sponsored R & D expenditure is even more concentrated).[12] Also, of all patents assigned to all corporations between 1946 and 1962 in the United States, resulting from corporate and government financed research, 35·7 per cent were owned by about 100 large corporations.[13] Since most of the patents in developing countries are foreign owned they reflect the concentration indices of their foreign origin. For example, in the Republic of Colombia of almost the total number of companies and individuals (foreign and national) who had patents on pharmaceuticals in 1970, less than 10 per cent (all foreign owned) controlled more than 60 per cent of the patents.[14] Also, from a sample of about 1,000 patents, mostly in the synthetic fibres and chemical industries, the concentration indices were the same as in the pharmaceuticals. The uneven concentration of patents in the hands of a few foreign firms mirrors the concentration of production in the respective industries or product-areas since patents define, sometimes quite exclusively, the firms that can produce (or import) certain goods.

'Once a company achieves dominance in an industry through patents, it creates the very condition for its perpetual control. If often provides the principal or only market for the patents of the independent inventor. It becomes the principal or only place of employment of the professional inventor and laborer in the field[15]. . . . In effect it commands the main stream of inventive thought and perpetuates its power through patents' [*Vaughan, 1948, p. 218*]. As an example of what occurs once patents enter in the field of business policy and how far the impact is from providing incentives to inventive activity, the following internal company memorandum is cited from Hartford Empire, the holding company which has dominated the glass bottle industry: '. . . Patent applications are being made so as to prevent the use or improvement of any existing or possible substitute of the bottle-making machine. . . . This . . . seeks to block competing devices which would lessen our income. . . . We now have a number of applications which were filed to definitely forestall the development of competing machines by others.'[16] Once monopoly privileges leave the world of the individual inventor (a romantic memory of the past which, however, still influences government policy) and enter the real world of multiple patent ownership by large corporations, the main function of patents is not to encourage inventive activity but to aid profit maximization through minimization of competitive forces.[17]

In conclusion, practically the totality of patents granted in developing countries are foreign owned, and their main function is clearly not to encourage inventive activity (domestic or even world wide) but to assist profit maximization of large transnational corporations. We pass now to the effects this has on the economies of developing countries.

PATENTS AND FOREIGN DIRECT INVESTMENT

In discussing the relationship between patents and direct foreign investment we will be subdividing our analysis into two parts:

(A) *Do patents obviate the need for investment, national or foreign?*
(B) Do patents substitute investment by patent holders for investment by non-patent holders?

(A) *Patents as a substitute for investment*

The more sophisticated arguments in support of the patent system accept the fact that most patents in developing countries are foreign owned and stress the importance of patents, not in their effects on inventive activity, but in the stimulus they provide to foreign investment. Through the patent process, the argument runs, developing countries are able to obtain patented and unpatented technology which would have been unavailable in the absence of adequate protection provided by patents. Thus, it is claimed that patents are a necessary condition for foreign investment. Although this argument appears repeatedly in patent literature, it lacks empirical verification. For example, an independent study undertaken for the U.S. government concluded: '. . . It would seem, therefore, that although investors and their patent attorneys may wish for better patent protection if it can be had, the nature of that protection is rarely a significant factor in the ultimate decision whether to invest' [*U.S. Senate, 1957, p. 17*].[18]

A growing body of theoretical and empirical investigations[19] on the complex motives for direct foreign investment indicate that the key motive (particularly in developing countries) stems from a defensive strategy [*Aharoni, 1966*]. It is to preserve markets that were once captured through exports and are subsequently threatened by competitors and/or by the import-substituting strategies of the host countries. In this context, patents, far from providing a stimulus to foreign investment, appear to be a critical factor in blocking investments, both foreign and national, in developing countries. As far as foreign investments are concerned, what more formidable means for countering these competitive threats that normally force a company to invest than the monopoly privileges which are given to the patent holder? The patent holder is the only enterprise who can produce *or* import the patented products in the (patent granting) country. Firms have openly stated that they take patents in other countries and suppress them (i.e. do not work them) as a way of 'stopping competition at its source' [*Penrose, 1951, p. 107*]. They export the patented products to markets where neither domestic nor other foreign firms can produce or export. 'Generally, restraints of this type [i.e. restraints on setting up new firms in developing countries] are exercised by means of patents taken out by a producing concern in one of the industrial countries, either for a particular type of machine or for a particular manufacturing process. The effect of such a patent is to prevent a firm in any country in which it is valid from using the machine or process in question except with the permission of the [patent] owner. . . . The patent owner often prefers to supply these markets from larger establishments in more advanced countries. . . . In the mercantilist era, such interests [opposing local manufacturing] might have enlisted the direct aid of government; at present it is chiefly by the use of economic weapons [such as patents] that they succeed in retarding local industrial development' [*United Nations, 1955, p. 27*]. An official report by the Indian government concluded that '. . . these patents are therefore taken not in the interests of the economy of the country granting the patent, or with a view to manufacturing them, but with the main object of protecting an export market from competition from manufacturers, particularly those in other parts of the world' [*Government of India, 1959, paras. 28–9*].

Patent holders achieve such objectives by the *acquisition of patents which are not used for production nor licensed for similar purposes to third parties in a given country.*[20] In the Republic of Colombia, out of a total of 3,513 patented processes or products examined (2,534 of which belonged to the pharmaceutical industry and the rest mainly to the textile and chemical industries) only 10 were actually produced in the country in 1970.[21] In Peru from a sample of 4,872 patents granted between 1960 and 1970 in major industrial subsectors (including electronics, textiles, machinery and equipment, chemicals, food processing, pharmaceuticals, fishing industry, metal processing, air transportation), only 54 were reported as exploited, i.e. only about 1·1 per cent of the total.[22] The motive for taking out patents in developing countries and leaving them unexploited is twofold: first it preserves secure import markets for large foreign corporations without necessitating foreign investment, and second it blocks potential competition from other close substitutes which can neither be produced nor imported. *Thus, patents granted by developing countries are not only almost all foreign-owned but they are almost totally unexploited.*[23]

A direct result of the patent system is the practice of cross-licensing of patents or patent pooling. This is a method whereby patent holders (mostly corporations) pool together their patents for competitive products and through explicit agreements reshuffle their monopoly privileges in order to divide world markets between themselves and avoid competition. 'Most of these cases involve some division of markets within Western Europe; a separation of the German from the French market, or the Italian from the German market, or similar compartmentalization. Typically, too, these arrangements separate the British market from the continent. . . . *The markets of the underdeveloped or non-producing areas are then allocated in accord with the respective bargaining power of the various producers*' [*U.S. Senate, 1957, p. 10*] (emphasis added).[24]

Thus, through the artificial restrictions they impose on the right to use or imitate certain key industrial techniques, patents prevent rather than foster investment (national or foreign) and preserve the markets of developing countries for imports sold under conditions of monopoly. We will be dealing with the resulting price effects later. Here we wish to stress that, perhaps the highest social cost of the patent system results from the restrictions they put on a country's opportunities to use its own and the rest of the world's available resources as it chooses.

(B) *Patents, ownership structure and forward linkage effects*

By controlling the production or even just the importation of patented products (or products manufactured by patented processes), the patent holder is able to restrict and direct the usage of such products to particular areas of activity in the patent-granting country. Furthermore, partly because of the absence of adequate (or any) anti-trust enforcement in developing countries, the patent holder can even determine the firms which can *use* the products or processes covered by patents.

Through these forward linkages, monopoly privileges associated with patented products can be extended into areas where patentability does not apply. For example, a firm which has patents for latex production will be the only one which can import latex and use it to produce other non-patentable rubber products, such as automobile tyres. Also, the owner of patented processes for producing certain active pharmaceutical substances can determine which firms can import and use such substances so as to 'produce', or even just package, antibiotics for final consumer usage.[25]

Extension of monopoly privileges to non-patented products through control over patented imports is also experienced in industrialized countries and has been prosecuted in their courts.[26] In developing countries lack of adequate legislation or even legal awareness of these matters leads to a definite monopoly control of non-patentable products through exploitation of the monopoly privileges extended by patented ones.

Since patents are almost totally foreign-owned in developing countries, they have significant effects on the ownership structure of the firms in certain industries. We will use the pharmaceutical industry as an example. This is an industry where patents play an exceptionally critical role. In Colombia 32 out of about 50 foreign firms control more than 74 per cent of a market in which there are also 139 nationally owned firms.[27] Out of 160 pharmaceutical laboratories that operate in Venezuela only 35 of them belong to Venezuelan capital. The foreign firms control more than 90 per cent of the pharmaceutical market.[28] There is similar percentage control by foreign firms in the Brazilian industry. Since the middle of the 1950s, Brazil has experienced a progressive denationalization of its industry in this sector, to a great extent because of the patent system that operated prior to the government decision in 1969 to suspend product *and* process patent grants in the pharmaceutical industry.[29] An analysis of the Brazilian market concluded that '. . . these large [foreign] groups achieve the consolidation of their privileged position through various means, starting from the patents. Due to patents and cross-licensing of patents . . . it is practically impossible for a developing nation to [expect] a reasonable independence from foreign sources. . . . We consider such a policy [suspension of patent grants] as the most important policy that our authorities could undertake; without it the pharmaceutical industry in Brazil will continue to be only a set of [foreign-owned] organizations that sell foreign products'.[30]

In developing countries patents serve as a powerful means for acquisition of nationally-owned companies by foreign concerns. (41 per cent of all U.S. investments in the Colombian chemico-pharmaceuticals industry, between 1958 and 1967 inclusive, took the form of acquisitions.[31] This constituted one of the highest percentage of acquisitions in all industrial subsectors in Colombia.) The usual procedure for such acquisitions is as follows: a foreign company, which has no subsidiary nor experience in a country, licenses the use of a group of patents to a nationally-owned company for a certain period of time. Once the market for the patented products has been tested and established (which includes the development of marketing channels) and the licensor has acquired experience in the various aspects of doing business in the country, the licences are suspended. The local firm is then 'convinced' to sell part or all of its stock to the foreign licensor. In this way, monopoly privileges derived from patents are instrumental in the displacement of local companies.

We can restate our main conclusions concerning the relationship between patents and direct foreign investments in developing countries. Patents are rarely a significant factor in promoting foreign investments. In most cases, patents guarantee secure export markets for the patent holders, and consequently are a major obstacle in creating the conditions that stimulate foreign investment. Where foreign investment takes place patents become one of the key instruments in the acquisition of local companies.

PATENTS AND NON-TRANSFER OF TECHNOLOGY

A mistake which is made repeatedly in the literature is to identify patents as a means or vehicle for technology transfers. Strictly speaking a patent is a legal document which gives an exclusive privilege to undertake production, make sales or import specified products or processes which meet certain legal requirements. In itself, therefore, the patent has no more to do with technology *transfer* proper, than for example a document which confirms the lease of a factory, or a legal paper which verifies the ownership of a house or, as an extreme case, a marriage certificate. Transfer (or non-transfer) of technology takes place not because of the patent itself but because other conditions exist (or do not exist) that create an incentive for productive activities (whether or not they are related to patents) which require know-how 'originated' elsewhere.

We shall now consider how patents regarded as legal instruments might influence the transfer of technology. Business executives and patent lawyers have consistently claimed that the protection offered by patents is an excellent stimulus for interchange of knowledge between enterprises. This is particularly true, it is claimed, where there is a possibility of cross-licensing. This argument does not go unchallenged. It is argued, for[32] example, that 'technology transfer' is in fact a rationalization for justifying cross-licensing activities, which have the main objective of market segmentation and monopoly control.[33] But whatever may be the arguments for the potentially positive effects of technology transfer through cross-licensing between industrialized countries, they are clearly inapplicable to developing countries. The developing countries find themselves excluded from such potential benfits simply because they own extremely few patents that they can cross-license with firms in the industrialized countries [*Freeman* et al., *1968*].

Thus, in order to evaluate the effects of patents on the flow of technology to developing countries we need to search in areas other than that of cross-licensing.

In analysing more than 300 contracts of technology transfer and patent licensing in the Andean Common Market, we rarely encountered cases where patent licensing existed alone. It practically always appears in combination jointly with contracts for the sale of know-how. Similar conclusions have been reached by other researchers [*e.g. Lightman, 1970, pp. 3–4*]. For example: '. . . transfers which *simply* provided . . . enterprises in developing countries with permission to exploit patents would very probably fail commercially' [*Cooper, 1970, p. 17*], emphasis added. Also, '. . . in the case where patented process technology is transferred, the unpatented components, which might appear as a "mere implementing appendage to the patent", are in fact the "more important and valuable component" of the transfer' [*Froes*, et al., *1963* cited in *Cooper, 1970, p. 5*].[34] The fact that patent licensing and technology commercialization contracts appear jointly does not necessarily imply any kind of complementarity between them: for example, the know-how which is transferred is not necessarily for the working of the patent itself. In fact, the licensing of patents and the sale of know-how have different functions in the transfer process. Technology commercialization contracts simply reflect the contractual terms in the sale of know-how. Patent licences, however, serve purposes which have more to do with the non-transfer of technology than with transfer:

(*a*) We have seen more than 98 per cent or 99 per cent of the patents in various sectors in Colombia and Peru were unexploited, i.e. the products or processes covered by such patents are not actually produced in these countries. Instead, some of the products are imported to be used in the production of non-patented (or non-patentable) products. Studies in progress in Chile, Bolivia and Ecuador suggest that a similar situation pertains to that in Peru and Colombia. Thus, patent licensing does not transfer the technology related to patents since the patented products are not produced and the patented processes are not themselves used in productive activities in the patent-granting *developing* country. Patents give an exclusive monopoly position in a national market and *by not being exploited*, they function so as to block the transfer of technology related to the patented products.

For example, although none of the Andean countries produces antibiotics and although the regional market justifies their production, efforts to import available and accessible technology from various sources around the world were frustrated by the patent holders. They preferred to supply the Andean countries with imports that they alone regulated.

As far as patents and the flow of non-patented technology is concerned the argument is parallel to that on the negative effects of patents on foreign investment. Technology, like private capital, is transferred to developing countries basically as a defence against the threatened loss of export markets for products. The closer a patent is to a final product the more it tends directly and indirectly to block transfer of non-patented technology since that product can be imported from abroad in a secure monopoly-controlled market.

In most cases a transnational corporation decides to sell technology to a developing country not because it has an assured monopoly position through patents, but because if it did not sell such technology some other corporations would have done it. If the non-patented know-how is 'captive', because it is secret or unknown to others, then patents do not offer additional security to the possessor, and are redundant. If, on the other hand, the non-patented technology is known to others (even though it is not known to the receiver in the host country) then one of the enterprises possessing that information may be induced to sell it to avoid other competitors doing so and thus replacing him. It is not security but insecurity of market control (otherwise known as 'competition') that stimulates transfer of technology and capital to developing nations. (Insecurity of market control is, of course, distinct from insecurity because of political instability, or uncertainties about ability to repatriate profits or royalties, which tend to have a retarding effect on the flows of capital and technology) '. . . The United States Antitrust Division of the Department of Justice has taken action against several international cartel arrangements [involving exclusive patent licensing agreements] . . . to which United States firms have been parties and almost invariably the exchange of technology has continued but on a less restrictive basis. It is apparently the experience of the U.S. Antitrust Division that while a large number of businessmen faced with prosecution of such arrangements have threatened to cut off all further contact with their foreign associates, few if any cases are known where this threat has been carried out' [*Penrose, 1951, p. 196*]. Similar conclusions were drawn in a Canadian Government report on the impediments to the transfer of technology related to products in electronics due to patents privileges in this area [*Commissioner, 1945, p. 22*, cited in *United Nations, 1953, p. 19*].

(*b*) Having indicated why patents tend to have a negative effect on the flow of technology we may now discuss why patent licensing co-exists with contracts for sale of know-how. The reasons for this coexistence are various. Patent licensing is one of the major reasons for the lack of diversity in the sources of supply of unpatented and patented technologies used by developing countries.[35] A patent holder grants a licence on condition that related bits of technical know-how will also be bought from him.[37] Also the patent holder, by virtue of his monopoly position, is able to tie the licence of the patent in a package form with a supply of capital (foreign direct investment) and/or the supply of intermediate products and capital goods. Returns to such tie-in arrangements can be much higher than the returns to technology sales (i.e. royalties). Thus, patents and the monopoly privileges that stem from them are the basis for the various types of 'tying' which accompany technology transfers to developing countries. It is for this reason that technology contracts and licence agreements are so often linked together. Finally, from the (legal) nature of patent privileges stems a type of 'ownership' non-existent in technology *per se* (nobody can 'own' information as long as it is not totally secret). From the monopoly power derived from such 'ownership' the technology supplier is able to impose conditions, in addition to tie-in arrangements, which would have been impossible in a more competitive market for technology. (These conditions, including restrictions on size and structure of production, price fixing, export restrictions, etc., will be dealt in the next part.) Hence, again, the coexistence between patent licensing and technology contracts.

Patent licences, which are so often considered vehicles of technology transfer, are in most cases nothing more than import permits (since most patents are non-exploited in developing countries) given under certain condi-

tions.[37] The state grants monopoly privileges to foreign patent holders and consequently national as well as foreign-owned firms need import permits, which they get in the form of patent licences from the foreign patent owners, so as to import the protected products. Once a government has decided to grant patents the ability to import products covered requires the permission of the patent holder who in turn gives it only under certain terms.

To summarize: since most patent licences given are in effect import permits; since the majority of patents are unexploited in developing countries; and since patents are a block to technology transfer and foreign investment, *patents are a means for limiting or restricting the flow of technology from the industrialized to developing nations.* '. . . Recent disclosures have abundantly documented this retarding tendency of patent reinforced monopoly in introducing commercially feasible inventions: in synthetic rubber, magnesium, methy-methacrilate plastics and dentures, tungsten carbide, higher potency pour-point depressants, fluorescent lighting, longer-life flashlight batteries, the repeater match, to name a few cases at random' [*Kahn, 1948, pp. 249–50*].

Patents are not only an impediment to technology transfer. They can also hinder possibilities for domestic research activities in developing countries. 'A single 17-year monopoly of a minor cog in that huge mechanism of interlocking processes and contributions which make up an advancing art can for 17 years seriously retard continued research' [*Kahn, 1940, p. 482*]. This is particularly important in view of the type of technological development and research needed in developing nations. Since a lot of such research should be directed to adaptation or modification of imported technology, the monopoly rights granted to foreigners by the existing patent system seriously blocks relevant domestic research activities. The technological infrastructure that might be created by imitating, absorbing and adapting foreign technology to domestic conditions is restricted by patents. After a 17- or 20-year period during which adaptive research is excluded, the technology covered by patents is probably obsolete and the search for adaptations becomes commercially unattractive.

From interviews undertaken in Italy in the course of Andean Common Market studies on technology transfer[38] it was concluded that the abolition of product *and* process patents law enabled them to undertake research to improve and amplify the uses of technologies which would have been inaccessible to them if patents had existed. Research and development activities have been directed to studying the know-how covered by patents in other countries with such success that Italy presently sells technology to the rest of the world. We conclude by citing the results of a Chilean study: '. . . in summary it can be said that the legal system in matters related to patents is in some way or another inhibiting local technological development' [*C.O.R.F.O. 1970, p. 13*].

RESTRICTIVE BUSINESS PRACTICES AND PATENTS

'The owner of the patent can specify what products his licensee is to manufacture, in what quantities they are to be produced, within what areas they are to be marketed, at what prices they are to be sold. The relation of patent-owner and licensee falls into a kind of feudal formula of Lord and vassal' [*Hamilton, 1945, p. 588*].

Patent licensing has provided classic examples of restrictive business practices and violation of antitrust or antimonopoly legislation. Such practices are also made possible by the terms on which technology sales take place as well as by ownership ties (e.g. a parent corporation can prohibit exports of products by its foreign subsidiary to a third country where another subsidiary exists), but as far as developing countries are concerned, patents seem to be a particularly important cause of restrictive practices. Much of technology actually imported by non-industrialized nations is widely known and can be acquired from various sources in the industrial world. When this type of technology is in question, intelligent national local entrepreneurs can exercise their negotiating power and avoid restrictions on the use to which the know-how is put. Patents, on the other hand, carry monopoly control even if the technology they cover is potentially accessible through various firms. The licensee's negotiating power is, in this case, seriously weakened. Thus, although the role and effects of patents have often been underplayed as compared to the non-patented know-how used in developing countries[39] it remains true that patents can constitute a serious cause of restrictive business practices.

Recent research in Colombia has indicated that a very high percentage of contracts for patent licensing and sale of technology include clauses which put limits on the licensee's business practices. For example, in those contracts in the textile, chemical and pharmaceutical sectors that contained relevant information, about 85 per cent explicitly prohibited exports by Colombian-owned firms and joint ventures [*Vaitsos, 1970b, pp. 20–1*]. Research undertaken in Chile, Peru, Ecuador and Bolivia has revealed equally severe and even more severe restrictive practices.[40] Some contracts which were limited simply to technology purchases or technical assistance without patents and which were negotiated basically by large locally-owned firms were exempt from restrictive practices. Contracts that included patents practically always included all the major types of restrictive clauses.

We proceed to list some types of restrictive business practices exercised through patent licenses, drawing examples from the rulings of the Supreme Court of the United States as well as lower court decisions in that country:

(1) *Price fixing*

(*a*) Prices charged by the licensee fixed by the licensor if only part of the product is covered by patent (U.S. *v.* General Electric Company (Carboloy), 80 F. Supp. 989, 79 USPQ 124, S.D. N.W. 1948) or if patent covers the process or machine and not the product produced (Barber–Colman Co. *v.* National Tool Co., 136 F. 2d. 339, 58 USPQ 2, 6th Cir. 1943).

(*b*) Fixing of resale price of patented product (U.S. *v.* Univis Lens Co., 316 U.S. 211, 53 USPQ 404, 1942).

(*c*) Fixing of licensee's prices as part of a mutual agreement among distributors of competing products (U.S. *v.* Masonite Corp. 316 U.S. 265, 53 USPQ 396, 1942).

(2) *Discriminatory rates*

Discriminatory royalties charged to different licensees (U.S. *v.* United Shoe Machinery Corp., 110 F. Supp. 295, D. Mass 1953) Alaska Federal District Court in Laitram *v.* King Crob, Inc.).

(3) *Tie-in arrangements*[41]

(*a*) (Standard Oil Co. of California *v*. Imoted States, 337 U.S. 293, 305–306): (Atlantic Refining Co. *v*. FTC, 381 U.S. 357, 1965); (FTC *v*. Motion Picture Advertising Serv. Co., 344 U.S. 392, 1953).

(*b*) Lower royalty rates when tie-in arrangements necessitate the purchase of other products (B.B. Chemical Co. *v*. Ellis, 314 U.S. 495, 52 USPQ 33) and (Barber Asphalt Corp. *v*. La Fera Grecco Constructing Co., 116 F. 2d 211, 47 USPQ1).

(*c*) Mandatory package licensing (Hazeltine Research, Inc. *v*. Zenith Radio Corp., 239 F. Supp. 51, 144 USPQ 381; N.D. III, 1965).

(*d*) Exclusive dealing patent-misuse cases (Preformed Line Products Co. *v*. Fanner Mfg. Co. 328 F. 2d 265, 140 USPQ 500, 1964) and (Tampa Electric Co. *v*. Nashville Co., 365 U.S. 320, 1961.

(4) *Territorial limits: domestic and export restrictions*

(*a*) Geographical and trade limitations for resale of patented products within domestic market. (Keeler *v*. Standard Folding Bed Co., 157 U.S. 659).

(*b*) Limitations in world markets. (U.S. *v*. Singer Mfg. Co., 374 U.S. 174, 137 USPQ 808, 1963). (U.S. *v*. Imperial Chemical Industries, Ltd., 100 F. Supp. 504, S.D. IXX 1951).

(5) '*Grant-back*' *covenants on improvements of patented know-how* (Hartford-Empire Co. *v*. U.S., 323, U.S. 386, 64 USPQ 18).

Restrictive business practices due to foreign patent licenses are found in both developing and industrialized countries in a very high percentage of patent contracts. For example, with the advent of the Second World War and the confiscation or exposure of 'enemy property' in the United States it was revealed by the Alien Property Custodian in the annual report for 1945 (p. 117) that about one-half of all contracts of patent licensing between U.S. corporations and firms belonging to countries at war with the U.S. at the time had restrictive clauses on market segmentation, levels of production and prices.

PATENTS AND TERMS OF TRADE

As we have said, monopoly privileges granted by patents are basically exercised through the creation of secure import markets for the patented products in developing countries. International legal procedures have institutionalized the relationship between patent privileges and import markets. For example, article 5, paragraph 1, of the Convention of the Paris Union for the Protection of Industrial Property effectively states that *importations* of patented products from other signatory countries constitute sufficient grounds not to 'entail forfeiture' of patents (through nonexploitation), in the patent granting importing country.[42]

It is self-evident that monopoly positions in import markets, created by patents or other means, will lead to an increase in the prices of the imported products, and consequently to unfavourable terms of trade for the importing countries. In the absence of patents the prices of the corresponding imported products would have been lower since competition from alternative supplies would have led to lower margins for the exporters.[43]

The following is a list of patented, mostly intermediate, pharmaceutical products imported by Colombia in 1968 and of the corresponding percentage of overpricing as compared to prices offered by various producers around the world not covered by patents in Colombia and/or in other countries. Preliminary research has indicated similar overpricing figures in Chile and Peru. (The prices of the non-patent holders were officially published by the Instituto de Comercio Exterior of the Colombian Government. The research undertaken in Colombia and the resulting policy rules, led the U.S. Government to set up new procedures for the pricing of pharmaceutical products purchased through A.I.D. loans in Latin America.)

TABLE 7

Name of product	*Imports at F.O.B. prices U.S.S.*	*% Over-pricing*	*Name of product*	*Imports at F.O.B. prices U.S.S.*	*% Over-pricing*
Terramycin anfotera	46,000	300	Tetracycline metaphosphate	28,750	998
Terramycin chlorhydrate	271,000	257	Neomycin sulphate base	5,300	40
Tetracycline base	278,120	80	Dexametasone alcohol	74,250	293
				9,980	43
				57,750	400
Dexametasone phosphatisodic	30,000	435	Chlordiasepoxide	62,625	5,647
Hexylresorcinol	15,675	117	Dehydroemetina Hel	20,480	55
Hydroclorotiazide	14,000	1,400	Prometazine	6,850	452
				2,800	636
Indomethacin	80,000	417	Ethionamide	9,000	875
Hydroxocobalamine	20,445	78	Chlorpromazine	26,760	733
				37,980	404
Cyanocobalamine	12,140	17	Metronidazol	54,690	521
	15,400	36		14,250	160
Erythromicin ethylsuccinate	970,000	140	Proclorperazine dimaleate	30,000	400
Erythromicin ethylsuccinate	176,000	100	Ciproheptadine chlorhydrate	140,000	260
	982,019	42			
Erythromicin estolate	220,448	176	Chlorpropamide	10,620	200
Metimazol	3,311	3,460	Guanethidine sulphate	57,500	360
Ampicillin oral	170,940	158	Methyl dopa	92,000	333
	98,000	70			
Ampicillin sodiciny	20,000	101.3	Chlortetracycline	62,250	400
N. Pinolidine Metilletracycline	224,412	686	Declonicine Hel.	116,000	74
			Chloramfenicol legogiro	58,125	300
D. Pantotenate of calcium	1,387	287	Chloramfenicol succinate	40,200	50
Substance of valium	66,500	6,584	Medroxyprogesterone	101,660	130
			Frusemide	102,442	1,075
			Tolbutamide	46,200	600
			Dipirone	148,155	141

Source: From unpublished material presented by the Colombian National Planning Department during the Second Meeting on Technology Transfer of the Andean Common Market, Lima, February 1971.

The influence of patents in worsening the terms of trade of developing countries stems not just from individual patent holders. Mainly, such influence is exerted through patent cartels achieved by cross-licensing arrangements. Monopoly privileges are thus attained for multiple patent holdings which block competition from close product substitutes or from alternative processes for the production of the same product or family of products. It is this cartelization of families of products which creates monopolies not just in the Chamberlinian sense (i.e. where each seller is a monopolist of his own product) but in control of the whole market. The division of world markets through patent cross-licensing has been substantiated in various studies [*Edwards*, et al., *1945; Terrill, 1946, pp. 745–50; Vaughan, 1948, p. 222; Stocking and Watkins, 1951, pp. 447–90; H.M.S.O., 1951*] and court cases[44] and has led to the conclusion that: '... this pattern has appeared in enough critical industries that it must be counted as a significant factor in world trade' [*U.S. Senate, 1957, p. 10*].[45] In one of the most comprehensive studies undertaken the following products or families of products were listed as subject to patent cartels through cross-licensing arrangements: acetic acid, activated carbon alkalis, aluminiums, cellophane, dyestuffs of various types, pharmaceuticals, magnesium, nickel, petroleum products, hydrogen peroxide, lead compounds, nitrogen photographic materials, sulphur, sulphuric acid, synthetic resins, titanium compounds, ammunition of specified types, electric cables, incandescent lamps, electrical apparatus, diesel engines, bottle glass, plate glass, tungsten carbide, linoleum, motion picture film and equipment, optical goods, radio equipment, rayon and rubber thread [*Hexner, 1946, pp. 184–401*, cited by *U.S. Senate, 1957, p. 10*].[46]

International market control[47] through patent cartels results in profit levels that would otherwise be unattainable for the corporations. As in the case of indirect taxes or tariffs '... those adversely affected are not so conscious of their loss as those who benefit are conscious of their gains.... What is a most important gain for the more powerful industrial countries is, on the other hand, one of the most important costs to the less industrialized importing countries' [*Penrose, 1951, p. 96*]. The economic effects of patents have been so much overshadowed by the complex legal treatments of the subject that governments in developing countries are in general unaware of the impact on their economies. 'Where an international cartel agreement can be built on patents the authority of the states granting the patents, far from being an obstacle with which to contend, is enlisted to enforce the policies and decisions of the cartel' [*Whittlesey, 1946, p. 80*].

ECONOMIC ARBITRARINESS IN THE LEGAL SYSTEM OF PATENTS

Relevance of the concept of 'novelty'. As we saw in previous pages, practically the totality of the patents granted by developing countries belong to foreign corporations or foreign nationals. The argument that monopoly privileges through patents are an incentive to local inventive activity becomes meaningless and with it falls one of the pillars of the legal rationale for patents. If monopoly privileges through patents are more related (positively or negatively) to direct foreign investment than to local inventiveness, should they not be evaluated and considered as part of the national codes on foreign investment, as are tax credits or ownership structure? If patents affect the terms of trade of the patent-granting importing countries should not they be viewed as part of the commercial policies of such countries, as are tariffs or exchange rates? It makes no economic sense to have a legal edifice built on the concept of 'novelty', when the origin and effects of patents on developing countries rest elsewhere. In the early history of patents (in German princedoms of the sixteenth century Saxony or further back in the regulations of the Venetian Republic in the 1470s) the concept of 'novelty' in discoveries by *local, individual* inventors may have merited specific economic consideration, but *what is the economic relevance of 'novelty' in the case of patents* granted by developing countries if such patents are owned largely by *foreign corporations*? (In 1970 after the Peruvian Government indicated its intention to bring the mining sector under national control there was a sudden influx of registration of foreign patents in this sector. Clearly no major discoveries, or at least not such large numbers, were realized in mining during that year. This again shows up the irrelevence of 'novelty' in patents registered in developing countries.) At present, the actual practice of patent laws in developing countries appears to be similar to the *patent monopolies* of early English patent history in the period of Elizabeth and James I. At that time patent monopolies were granted so as 'to control the industry' (nowadays the case will apply to foreign control of industry), so as 'to reward favourites' (nowadays the transnational corporations and the local patent lawyers) and so as 'to secure loyalty to the Crown' (nowadays known as 'investment climate').[49]

This suggests that modern legislation on patents should be based on a real perception of their underlying economic role rather than the abstract notions of 'novelty'. As matters stand: '. . . The patent system in its present form is a highly artificial creation emerging from a process of legislation in which the role of the pressure groups and muddled thinking has been unfortunately only too prominent. . . .' [*Robbins, 1939, p. 73*, cited by *Penrose, 1951, p. 40*].[50]

RELEVANCE OF CONCEDING MONOPOLY PRIVILEGES TO AN 'INVENTOR'

The patent system makes certain 'economic assumptions' which may well be inapplicable in the technological complexity of a modern economy. '. . . The most important—and most fallacious—assumption is its individualistic conception of the process in invention. In fact, invention is a group process, the individual contributions being relatively minor. In modern industrial research this is particularly so' [*Kahn, 1940, p. 475*].[51] Serious economic questions arise from the extension of monopoly privileges to *a* company or *an* individual in a particular product or process given the continuous interdependence that exists in technological advancement. Furthermore, it is economically arbitrary to grant patents to a particular company and to exclude others when independent research often leads to almost simultaneous discoveries by several. The legal concept of 'priority' of registration at a patent office could thus become highly discriminatory. Ogburn lists 148 important discoveries made independently and almost simultaneously by two or more persons or companies. He suggests that the list is in reality almost endless [*Ogburn, pp. 90–102*]. In modern times of corporate multiple-patent holdings, private arrangements between corporations and out-of-court settlements often become the arbitrary means of defining priority of invention.[52] Relative bargaining power, financial strength in sustaining long legal battles or possible collusion between members of cartels appear much more to be the relevant factors in the actual ascription of monopoly privileges for innovations.[53]

Because of the very high cost and the expertise required for an efficient patent office,[54] the developing countries are in effect totally dependent on the decisions of the patent offices in the industrialized world in determining who should be granted monopoly privileges. Countries that have 'prior examinations' for the evaluation of patents find themselves with immense backlogs. For example, Brazil in 1970 had nearly 400,000 pending patent applications.[55] Countries without 'prior examination'[56] grant patents on the fulfillment of simple legal requirements on the implicit assumption that evaluation of 'novelty' and of other technical requirements has been undertaken in the country of 'origin'.[57] Thus, developing countries grant monopoly privileges in their own markets, to a great extent depending on decisions on 'novelty' made elsewhere. Such decisions involve a high degree of economic arbitrariness. For example, '. . . in 89 per cent of all patent cases which were brought up to Courts of Appeals and the Supreme Court of the U.S. during 1941–1945, the patents involved were declared invalid' [*Cullen, 1945, pp. 903–4*].[58] Of course, the invalidation of a patent in the country of its original registration does not mean its invalidation in other countries according to the principle of 'independence of patents' prescribed by Article 4 of the Paris Convention.[59] Thus developing countries, given the inefficiency of their patent offices, can continue under present legislations to grant monopoly privileges to products or processes which have been declared as non-patentable in their country of 'origin'.

Similar economic arbitrariness underlies all major principles of the patent system and the legislation that covers it in developing countries. We conclude this chapter with the reference '. . . that it is life and not logic that gives birth to patent law; . . . a system of legislation that has developed through and by accidents and incidents of the history of a nation has such a claim to the sympathy of that nation that no demonstration, no matter how complete, of the absurdity or even the bad effects of the system will serve to weaken this attachment' [*Ranton, p. 491*].

CONCLUSION

Most economic policies involve certain trade-offs. For example, tariffs protect domestic industries (infant or otherwise) but at the same time they perpetuate inefficiency and lead to a redistribution of income from consumers to producers. Tax incentives provide a stimulus to new investments but at the same time they involve fiscal losses. Thus one rarely encounters an economic policy which has so many predominantly negative effects as does the patent system in developing countries, without apparently any significant benefits.

Practically all patents in developing countries are foreign-owned and as such they do not have any direct relationship with domestic inventive activity. Since most of these patents are owned by large foreign corporations their implications are basically directed towards the overall business policies of these firms and patents are aimed at market segmentation through cross-licensing and monopoly control. It is also reasonable to conclude that patents granted by developing countries have no effect on the R & D activities of the large transnational corporations. Hence, practically all patents registered in developing countries have no direct causal relationship whatsoever with inventive activity, domestic or foreign.

The monopoly privileges enjoyed through patents constitute one of the main factors that block the market conditions which stimulate foreign investments in developing countries. When foreign investments take place patents serve as an effective vehicle for the acquisition of nationally-owned firms. Most of the patents granted by developing countries are not exploited in the sense that the patented processes or products are not themselves produced in the patent-granting developing country. Instead, secure import markets—achieved through legal barriers—enable the foreign patent holders to charge prices higher than they could expect under more competitive conditions in the absence of patents. This in turn deteriorates the terms of trade of the importing developing countries.

Far from constituting a vehicle of technology transfer, patents appear to hinder the flow of technology to developing countries as well as to restrict local technological advance through imitation and adaptation. The function of patents in contracts of technology sale appear to be more related to the achievement of restrictive business practices. Such practices, most probably, would not have been achieved under a more competitive technology market or under the application of antimonopoly legislation without the limitations imposed by patents. For developing countries the licensing of patents constitutes granting import permits under certain restrictive conditions. Finally, the legal edifice of the present patent system in developing countries is built upon arbitrary economic foundations which negatively affect the interests of the countries themselves. A legal concept of 'ownership' has been developed for ideas or innovations, exercised through artificially-created monopoly privileges. One is left uncertain where the line of 'ownership' should be drawn or why such monopoly privileges should exist in technological innovations and not in scientific inventions. Why should 'ownership' cum monopoly privileges exist in technology alone and not in other areas of creative endeavour such as in various forms of communications, in marketing, in scheduling and storing techniques, or in new financial practices? 'Ownership' and monopoly control were practised in previous ages in rivers, bridges, roads, etc., with significant retarding effects on the economies involved, and were subsequently abolished.

The patent system and the laws that cover it have grown through the continuous pressure of concentrated interest groups facing other producers or consumers with 'diluted' interests. But for the government of a developing country the question still arises: why should patents exist in developing nations?

NOTES

1. Some notable exceptions exist such as the work of Edith Penrose [*1951*].
2. For example, the specification as to who carries the burden of proof of an alleged infringement (the patent holder or the alleged infringer) could determine whether patent privileges and protection really exist or whether they only constitute inapplicable legal principles.
3. See, for example, B.I.R.P.I. [*1965, p. 17*].
4. For some economic implications of 'inexhaustibility' of usage of inventions and

the effects of a bargaining structure rather than a competitive market-price system see Vaitsos [*1970a, pp. 17–21.*]

5. Reference is made here to the Convention of the Paris Union for the Protection of Industrial Property and the effect that it and its Secretariat (B.I.R.P.I.) have had on the international patent system and its regulations.

6. See declaration by Brazilian delegation in 'Informe Final de la Primera Reunión de Directores de Oficinas Nacionales de Marcas y Patentes' ALALC/MAP I/Informe, May 1969, pp. 8–9.

7. For similar conclusions see Penrose [*1951, p. 233*].

8. Under the auspices of the O.E.C.D., the European Industrial Research Management Association was established to promote research activities in Europe. A major problem investigated refers to the obstacles imposed by European patent laws on R & D activities and technological advancement. For reference, on the lack of monopoly privileges in other fields of creative inventive activity, see Vaughan [*1948, p. 230*].

9. The document concluded that 'individual inventors are scarcely integrated in the local industrial system' [*Chudnovsky and Katz, 1970, p. 36*].

10. In terms of industry breakdown the following share of patents was registered in Chile in 1967.

11. The percentage of Argentinians in the total number of patents owned by individuals in 1968 was about 75 per cent.

12. For concentration effects of licence policy from U.S. government financed research see the testimony of Leontief (*op. cit.*).

13. United States Patent Office data analysed by Watson and Holman [*1967, p. 377*].

14. Deduced from data collected by Timoleón López and F. Castaño from the Colombia Industrial Property Office for the studies on technology transfer of the Andean Common Market.

15. In research and interviews undertaken in Peru for the Andean Common Market Study on technology transfer observations were made on Peruvian professionals working for patent-holding subsidiaries of foreign pharmaceutical firms in related fields of tropical diseases. Whatever discoveries were made by these subsidiaries with the participation of local professionals were transferred to the foreign parent corporations. Similar cases were cited for Brazil.

16. Memorandum was presented by Commissioner Coe in 'Investigation of Concentration of Economic Power' part three, p. 842, as cited by Kahn [*1940, p. 485*].

17. One of the industries where patents constitute a critical factor is pharmaceuticals where patents have become the chief cause of a cartelized world-wide market control with one of the highest profitability rates. The Till report, one of the main government documents presented during the Kefauver hearings in the U.S. Senate, gave evidence about the efforts of the pharmaceutical firms '. . . to preserve the *foundation of their prosperity, the patent system*, by means of which they were able to control the market . . .' [*Harris, 1964, p. 24*] (emphasis added).

18. U.S. Senate [*1957, p. 17*]. The study utilized various empirical references among which the following:

The National Industrial Conference Board undertook a survey on 107 U.S. companies which had very significant foreign investments around the world. The purpose was to determine the various obstacles that limit or prevent foreign investment. A series of obstacles were identified which ranged 'from the problem of export quotas to that of inadequate sanitary facilities'. In the 382 pages, which presented the findings of this research, patent protection or lack of it were not mentioned even once as affecting decisions on foreign investment (see N.I.C.B. [*op. cit.*]). Similarly, a survey undertaken by the United States Department of Commerce, involving 247 overseas investors on factors that negatively affect foreign investment does not mention the issue of patents either (see U.S. Department of Commerce [*1954*]).

As an indication of the diverse factors that prompt direct foreign investment one can cite the Italian case where heavy foreign investment took place in the pharmaceutical industry *precisely because of the complete lack of patents in that industry in Italy.*

19. See various studies related to the so-called 'product cycle' theory.

20. The present 'corrective actions' suggested by the existing international legislation (e.g. article 5 of the Paris Union) have proved of practically zero operational value as far as developing countries are concerned. For some statistical evidence see United Nations [*1964*].

21. From unpublished data collected in Colombia for the studies on technology transfer in the Andean Common Market. Part of the unused patents cover processes

or products that could have been introduced in the Colombian market by competitors. Another part includes cases that were possibly undertaken to assure complete legal coverage of monopoly privileges without necessarily representing immediate threat of competitive product introduction.

22. From research undertaken in the Peruvian Ministry of Industries and Commerce under the direction of Cesar Suito for the Andean Studies on Technology Transfer.

23. Non-exploitation of patents has preoccupied industrialized countries as well, although to a lesser degree. 'Perhaps the most important group of suppressed patents consists of the hundreds taken out in the United States by foreigners solely for the purpose of reserving this country as a market for their inventions manufactured abroad' [*Vaughan, 1948, p. 217*]. Also during the parliamentary debates in England in 1907 the issue was raised that half of the patents granted by that country belonged to foreigners. Most of them were suppressed and through imports of the patented products many British industries were 'completely wiped out'. (House of Commons, *Parliamentary Debates*, March 19, 1907, col. 684, as cited by Penrose [*1951, p. 140*].)

24. Also, '. . . the International Patent Convention—and the practices under it—are basic to the whole structure of the present . . . policy of restrictions on economic development in less industrialized countries and the monopolization or cartelization in the more industrialized ones' [*Kronstein and Till, 1947, p. 766*].

25. Vitamin tablets and hormone injections are clearly non-patentable. Yet the active and other substances used as intermediate products for them are. The firms that own or are licensed with the patents for these intermediate products or their processes are generally the only ones that can 'produce' or package the final products since they are the only ones that have access to the importation of the intermediate products.

26. For the U.S. experience see Marcoid Corp. *v.* Mid-Continent Investment Co., 320 U.S. 661; Ethyl Gasoline Corporation *v.* United States, 309 U.S. 436; Leitch Mfg. Co. *v.* Barber Co., 302 U.S. 458; Morton Salt Co. *v.* Suppiger; 314 U.S. 488; Carbide Corporation *v.* American Patents Corp., 283 U.S. 27, etc.

27. It is of interest to note that one rarely encounters other industries where such a large number of foreign firms operate in a developing country, with oligopoly or monopolistic positions in their particular product lines stemming basically from the effects of patents. Aside from monopoly considerations there are serious effects resulting from the compartmentalization of the market. From direct interviews in Colombia it was estimated that on the average the industry utilizes less than 50 per cent of one shift of its installed capacity. National firms indicated that in the absence of patents they could process closely to 90 per cent of the different products in the market thus increasing their present production by about 300 per cent without significant changes in presently installed capacity.

28. See *El Mundo*, Caracas, April 11 and May 9, 1970.

29. Percentage participation of foreign capital in the Brazilian pharmaceuticals market:

1957	66·0	1961	85·7
1958	70·0	1962	88·5
1959	75·0	1963	90·0
1960	80·4		

Source: Pacheco [*1968, p. 44*].

30. Gazeta, 'Remedios No Brasil, Custos e Producao', Sao Paulo, July 27, p. 23 (my translation from Portuguese).

31. See data of Vaupel and Curhan [*1969*] as cited and analysed by Wionczek [*1970, p. 10*].

32. 'Independent research accompanied by market competition among well-financed producers . . . could conceivably be conductive to faster technological development than pooled research without such competition.' U.S. Senate [*1957, p. 22*], opinion presented by R. Vernon.

33. For bibliography giving references to opposing viewpoints see Stocking and Watkins [*1951*].

34. See also Lachman [*1965*].

35. Another principal reason, of course, is the lack of sufficient knowledge on the part of the purchaser of the alternative availabilities in the technology market. For a conceptual analysis on this point see Vaitsos [*1970a, pp. 16–17*].

36. For a discussion of this and other related points see Charles Cooper [*1970, pp. 17–19*].

37. For example, from ten contracts submitted by transnational corporations to the U.S. Senate Subcommittee on Antitrust and Antimonopoly, nine referred to licensees who had produced and sold the related products prior to the licensing agreement. Furthermore, five of these recipient companies, as the contracts indicated, had developed their own individual processes of fabrication. The licensing agreements were signed since the products involved were patented and the litigations were in process concerning importation and production of patented intermediate products. (For complete text of these contracts see U.S. Senate [*1959–60, p. 16031* et seq.].)

38. The interviews were undertaken by a functionary of the Colombian Planning Department in December of 1969.

39. For such an analysis see United Nations [*1964*].

40. These studies were instrumental in getting acceptance of a proposal for some quite unique legislation which marked the beginning of a legal awareness on anti-trust matters by the Andean countries. See Decision 24 of the CACM [*1970, arts. 20 and 25*].

41. For the *per se* illegality of tie-in arrangements under the Clayton and Sherman Acts see Turner [*1958*].

42. It is really astonishing that the developing countries which are members of the Paris Convention could ever accept such a provision.

43. In 1970 the German-owned firm that controls the patent on indometacine sold to the Colombian Government for Col. pesos 700/thousand units. A North American firm without access to the patent offered the same product for Col. pesos 133·50/thousand units. The Colombian Government was forced to buy from the German firm through a court decision based on the patent coverage. (See *El Tiempo*, June 2, 1970.) The same issue was cited in Ecuador where the North American firm won the case due to a more liberal patent law in that country.

Also, 'the "lavish margins" given to importers and distributors of Dutch half-watt lamps was complained of by the Standing Committee on the Investigation of Prices and Trusts under the Profiteering Acts in Great Britain in 1920. In spite of the fact that Dutch lamps were in fact manufactured much more cheaply than the British, higher prices in Britain were made possible because only three firms were licensed to import lamps'. See p. 11 of the Report of this Committee in the House of Commons Sessional Papers volume 28 (1920) as cited by Penrose [*1951, p. 95*].

44. U.S. *v.* Imperial Chemical Industries Ltd. 100 F. Supp. 753 (D.N.J. 1949): U.S. *v.* National Lead Co. 63 F. Supp. 513 (S.D.N.Y. 1945); etc.

45. Also '. . . this patent-owing and leasing system exists in numerous industries and apparently is now the most effective scheme for restraining trade' [*Vaughan, 1948, p. 219*].

46. Also Vaughan lists the following products as subject to international patent cartel arrangements, as mentioned by the above-cited U.S. Senate study: beryllium, latex, aviation gas, aluminium, titanium, tungsten carbide, fluorescent lamps, coated abrasives, chemicals, magnesium, tempered glass, lecithin, fluid-filled cables, duplicating machines and stencils, gas-refrigeration equipment, screws and drivers, glass fibres, alloys and dentures, optical instruments, nitrates, automotive and aviation parts, matches, electric precipitators, storage batteries, plastics, television, photoprint materials, stud-welding equipment, water-conditioning apparatus, seamless-hosiery machinery, electric generators and equipment, prismatic glass, hydraulic brakes, dental powder, wiredrawing machinery, gear-cutting tools, railroad-car couplers [*Vaughan, 1925, pp. 54–59, 94–99, 145–165*]. See also Vaughan [*1948, pp.* 219, *222*].

47. The Kefauver hearings in the U.S. Senate led to the description of the following market structures and controls in the pharmaceutical industry in the 1950s.

Examples of monopoly

(1) Chlorotetracycline: American Cyanamid followed for several years since 1948 a policy of not granting licensing agreements to third parties in various countries.
(2) Chloramfenicol: Parke Davis; no licensing offered during the same period.
(3) Oxitetracycline: Pfizer Co.; no licensing offered during the same period.
(4) Tolbutamide: Upjohn Co. with exclusivity in U.S. market.

Examples of duopoly

(1) Meprobamato: Carter Co. and American House Products have kept prices in the U.S. practically the same for that product between them. Duopoly structure was preserved in the international market for considerable periods of time. The substantial profits achieved from this duopoly were discussed in the Hearings pt 16, pp. 9153–9157.

Examples of oligopoly

(1) Various products derived from tetracycline which initially was fabricated and sold internationally by five companies. Competition was basically in the form of brand names.

(Source: U.S. Senate, Anti-trust and Anti-monopoly Subcommittee 'Study on Price Fixing in the Pharmaceutical Industry', 87th Congress, 1st Session, 1961, Spanish translation pp. 50–57).

The effects of market controls through patents on prices are indicative in the three generations of antibiotics:

(1) *Penicillin:* The technological advancement (1944) by the U.S Department of Agriculture laboratory in Peoria for commercial exploitation led to patents available to any producer without charge. Strong competitive forces resulted in continuous price decreases.

(2) *Streptomycin:* It was patented and licensed to many on a royalty-payment basis. This led to higher gross margins than penicillin but still competitive forces were relatively strong.

(3) *Chloranphenicol, Oxitetracycline:* They were patented and initially not licensed to third parties. Gross margins were among the highest in the pharmaceutical industry and considerably higher than average gross margins in other industries.

Source: Harris [*1964, pp. 22–24*].

48. Also 'There is an obvious reason why patents were cherished as a framework for private market controls. In addition to corporate controls and trademarks, patents were the only tie "*legitimate*" all over the world which could be used to bind entrepreneurs in marketing controls' [*Hexner, 1946, p. 73*] emphasis added.

49. For a brief description of patent history see Penrose [*1951, pp. 1–18*], and also Ladas [*1930*]. It is of interest to note that the first law on patents in England appeared under the Statute of *Monopolies* of 1623.

50. For foreign interference in national patent systems see:

(*a*) Pressures of U.S. firms on the U.S. State Department to exert influence on other governments so as to adopt stricter patent laws for the protection of U.S. products. (Information on company policies and their pressures on the U.S. State Department appeared in Senate Hearing; see U.S. Senate [*1959–60, p. 59*].

(*b*) Influence of German chemical industry on its government to use the German-Swiss tariff negotiations of 1904 as a leverage for the adoption of a new, stricter patent law in Switzerland (see Stuber [*pp. 26 ff.*]). Strong pressures were also exerted against the Swiss government by B.I.R.P.I. on the issue of the location of its offices in Geneva.

(c) Influence of German chemical industry on its government to exert diplomatic pressure and apply retrospectively methods against English patent holders in Germany 'so as to avoid the passage of the compulsory working provision of the English patent law' in 1908. See Hoffman [*1933, p. 101*].

(*d*) The effect of making U.S. participation in the 1873 international exhibition in Vienna conditional on the adoption of stricter patent laws by the Austro-Hungarian Government. 'Under such laws, it would be well for the Austrians to consider whether the invitation to nations is not likely to be met with contumelious refusal' [*American Ministry in Vienna, 1872, p. 84*, as cited by *Penrose, 1951, p. 45*].

(*e*) Efforts in 1970 of several embassies of industrialized countries to force Colombia's adhesion to the Paris Convention. The Council of Political Economy of Colombia vetoed the decision by the Senate (which found itself under considerable lobbying) for such adhesion.

(*f*) Influence on the content of the Treaty of Washington of 1907 (called the General Treaty of 'Peace and Amity'). It was signed by Central American countries after several years of serious fighting and disputes. In the Treaty there were included provisions regarding the treatment of patent privileges. See Penrose [*1951, p. 127*].

(*g*) Influence of B.I.R.P.I. 'Many of ... the developing countries do not feel interested enough to become members [of the Paris Union] inasmuch as they believe that they would assume more obligations in exchange for benefits of a meagre character. ... It is necessary that a propaganda should be instituted in these countries, to convince them of the advantages afforded to them by the Union. This would seem to be the business of the Bureau' [B.I.R.P.I.] [*Ladas, 1930, p. 808*].

(*h*) Influence of B.I.R.P.I. in convincing the signatory countries of the Afromalagasy Accord to adhere to the Paris Convention. See United Nations [*1964*].

51. Kahn continues to note that '. . . it is little short of absurdity to call any one of the interrelated units *the* invention, and its creator *the* inventor. The man [or company] who brought to a certain stage of fruition the efforts of myriad predecessors and who laid the groundwork for the efforts of myriad successors, and whom therefore we call the inventor, may have made a great contribution. But seen in its proper setting, that contribution is less than cataclysmic' [*Kahn, 1940, p. 479*].

52. See, for example, settlement of priority for the discovery of benzatinic penicillin between Wyeth, Pfizer, Lilly and Bristol in 1952. Also see for the discovery of prednisone, a similar case between four other companies.

53. '. . . From so costly, vexatious and uncertain a campaign at law all parties are likely to shrink. A settlement is usually made by which each of the strong issues licenses to all the others under its patents. The weak have to accept what is offered. . . .' [*Hamilton, 1945, p. 590.*]

54. In 1942 the U.S. Patent Office, despite its considerable funds and technical staff, had more than 100,000 applications and amendments awaiting action by the examiners. The average period of time necessary for passing an application through all the requirements of the Patent Office was six years [*Davis, 1948, p. 240*].

55. *Business Latin America*, March 12, 1970, p. 86. The backlog of patent examination does not necessarily mean effective lack of monopoly privileges. National or other foreign firms do not mount production facilities out of fear that once patents are granted they will be faced with potentially unusable resources.

56. See article 9, law 31 of the 1925 Act of the Republic of Colombia.

57. In these cases the state does not guarantee the 'novelty' of a process or product but it grants patents on the assumption that 'novelty' exists unless challenged and proven to the contrary. As a Latin American businessman we interviewed indicated, '. . . how can I challenge the novelty of a patented product by foreign corporation X when I depend on such a corporation for the license of other products and for the use of such foreign trademarks as Alka Seltzer, etc.? Anyhow the costs of challenge in the Courts are too high to be worth the battle. . . .'

58. Also in 1963 the U.S. Supreme Court came across a settlement of interferences of patents which it found '. . . to have been executed for the purpose of facilitating the issuance of a patent with broad coverage and strengthening the two inventors' efforts to stifle competition from other manufacturers' [*Bureau of National Affairs, 1966, p. 11*]. Justice White in a concurring opinion concluded that 'the purpose of the settlement was to prevent prior art from coming to the Patent Office's attention'; see U.S. *v.* Singer Mfg. Co., 374 U.S. 174, 137 USPQ 208 (pp. A–12, X–30, ATRR No. 101, 6/18/63).

59. For example, in the first months of 1970 the United States Justice Department asked for cancellation of the 'fraudulently procured' ampicillin patent and the invalidation of ampicillin trihydrate patents. The 1968 world wide sales of this semi-synthetic penicillin only by one company and its licensees were approximately U.S.$170 mil. Patents for the 'fraudulently procured' ampicillin were taken in more than 60 countries (see F.D.C. [*1970*]). Despite the action taken by the U.S. Government, the Patent Offices of the Andean Countries as well as those of other nations continued to grant monopoly privileges to these products if previously patented.

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